AI CAREERS FOR WOMEN
LINKEDIN FELLOWSHIP

[Employee payroll data cleaning and financial analysis]

Project Domain

[e.g., Excel / Power BI with Copilot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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ABSTRACT OF THE PROJECT

This project is developed as part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, which aims to provide students with practical exposure to Artificial Intelligence, Data Analytics, and Business Intelligence tools. The main objective of this project is to analyze and improve the quality of employee payroll data using Microsoft Excel and AI-assisted tools.

Payroll management is an important function in every organization, as it involves processing employee salaries, bonuses, overtime payments, deductions, and calculating the final net salary. However, payroll datasets often contain issues such as duplicate records, missing values, inconsistent department names, and incorrect salary calculations. These issues can lead to inaccurate financial reporting and inefficient decision-making.

In this project, a payroll dataset was collected and explored to understand its structure and identify potential data quality issues. Various data cleaning techniques were applied in Microsoft Excel to remove duplicates, handle missing values, correct formatting errors, and validate salary calculations. Important payroll performance indicators such as Total Payroll Cost, Average Salary, Highest and Lowest Salary, Department-wise Payroll Distribution, Bonus Percentage, and Deduction Ratio were calculated using Excel formulas and Pivot Tables.

Additionally, an AI chatbot tool was used to assist in validating formulas, generating analytical insights, and documenting the project findings. The final outcome of this project is a clean and structured payroll dataset along with an analytical report that provides meaningful insights for HR and finance teams. This project demonstrates how combining traditional data analysis tools with AI assistance can improve data accuracy, efficiency, and decision-making in organizational payroll management.

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CHAPTER 1

Introduction

Payroll management is an essential function in every organization as it ensures that employees are paid accurately and on time. It involves managing employee salary records, bonuses, overtime payments, deductions, and calculating the final net salary. As organizations grow, payroll data becomes larger and more complex, making it difficult to manage and analyze manually.

Payroll datasets may contain issues such as duplicate records, missing values, incorrect salary calculations, and inconsistent data formats. These problems can reduce data accuracy and affect financial reporting and decision-making.

This project focuses on cleaning and analyzing an employee payroll dataset using Microsoft Excel. Various data cleaning techniques and analytical methods are applied to ensure data accuracy and generate useful insights. AI tools are also used to assist in validating formulas and generating analytical summaries. The project demonstrates how combining data analysis with AI tools can improve efficiency and support better decision-making in payroll management.

1.1 Problem Statement

➤ Large organizations manage extensive employee payroll datasets that include information such as employee ID, department, gross salary, bonuses, deductions, and net salary. These large datasets often become difficult to manage and may contain data quality issues such as duplicate records, missing values, inconsistent department names, and incorrect data formats, which reduce the reliability of the data.

➤ Payroll data accuracy is extremely important because it directly affects salary payments and financial reporting within an organization. Errors in payroll records can lead to incorrect salary calculations, financial losses, and dissatisfaction among employees, making it essential to maintain clean and validated payroll data.

➤ Several stakeholders are affected by payroll data issues, including HR departments, finance teams, organizational management, and employees. HR and finance teams rely on accurate payroll information for salary processing, budgeting, and financial planning, while employees depend on correct payroll records for fair compensation.

➤ In many organizations, payroll datasets are still reviewed manually using spreadsheets, which can be time-consuming and prone to human error. Therefore, this project focuses on applying structured data cleaning, validation techniques, and analytical methods using Microsoft Excel and AI-assisted tools to improve payroll data accuracy and generate meaningful insights.

1.2 Motivation

➤ This project is meaningful because it provides practical experience in working with real-world organizational data. Analyzing an employee payroll dataset helps develop important skills such as data cleaning, validation, and structured data analysis using Microsoft Excel.

➤ Payroll data analysis has strong real-world applications in organizations where HR and finance departments rely on accurate salary records for payroll processing, budgeting, and financial planning. A clean and structured payroll dataset helps organizations maintain reliable financial information.

➤ Improving payroll data accuracy can help organizations reduce errors in salary calculations, improve transparency in payroll management, and support better financial and operational decision-making.

➤ The project also reflects modern industry trends where Artificial Intelligence tools are used alongside traditional data analysis methods. AI tools can assist in validating formulas, generating insights, and supporting report preparation, making the analytical process faster and more efficient.

1.3 Objectives

➤ To collect and understand the structure of the employee payroll dataset, including fields such as Employee ID, Department, Gross Salary, Bonus, Deductions, and Net Salary.

➤ To clean and preprocess the payroll dataset by removing duplicate records, handling missing values, correcting data formats, and standardizing department names to ensure data accuracy and consistency.

➤ To validate payroll calculations by checking whether the Net Salary is correctly calculated based on Gross Salary and Deductions and identifying any inconsistencies in the dataset.

➤ To perform data analysis using Microsoft Excel by applying formulas such as SUM, AVERAGE, MAX, MIN, COUNTIF, and SUMIF, and by using Pivot Tables to generate meaningful payroll insights.

➤ To calculate key payroll performance indicators (KPIs) such as Total Payroll Cost, Average Salary, Highest and Lowest Salary, and Department-wise Payroll Distribution.

➤ To utilize AI tools such as AI chatbots to assist in validating formulas, generating analytical insights, and supporting the preparation of structured project documentation.

➤ To analyze department-wise payroll distribution and identify patterns in salary allocation, bonuses, and deductions in order to generate meaningful insights that can support HR and financial decision-making within the organization.

1.4 Scope of the Project

➤ This project focuses on analyzing an employee payroll dataset that contains information such as employee ID, department, gross salary, bonuses, deductions, and net salary. The study aims to improve the accuracy and reliability of payroll data through structured data cleaning and validation.

➤ The tools used in this project are primarily Microsoft Excel for data cleaning, validation, and analysis, along with AI chatbot assistance for formula validation, insight generation, and supporting project documentation.

➤ The project mainly focuses on tasks such as identifying duplicate records, handling missing values, correcting data formats, validating salary calculations, and generating payroll insights such as total payroll cost, average salary, and department-wise payroll distribution.

➤ The scope of the project is limited to data cleaning, payroll analysis, and AI-assisted reporting using the given dataset. It does not include real-time payroll systems, machine learning models, or integration with live organizational databases.

➤ This project is conducted for academic and learning purposes as part of the AICW Fellowship Program and demonstrates how data analytics and AI tools can be applied to improve the quality and understanding of payroll data.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

In many organizations, payroll management systems are used to maintain and process employee salary records. These systems handle important information such as employee identification, department details, gross salary, bonuses, deductions, and net salary. Several studies and industry practices highlight the importance of maintaining accurate payroll data to ensure correct salary payments, financial transparency, and efficient workforce management.

Previous work in the field of data analytics emphasizes the importance of data cleaning before performing any analysis. Researchers have shown that large organizational datasets often contain issues such as duplicate records, missing values, inconsistent naming conventions, and incorrect data formats. Data cleaning techniques such as removing duplicates, correcting formatting errors, and validating calculations are essential steps to improve data quality and reliability.

Spreadsheet tools such as Microsoft Excel are widely used in organizations for managing and analyzing payroll datasets. Excel provides various functions such as filtering, sorting, conditional formatting, formulas, and pivot tables that help analysts identify inconsistencies and perform structured data analysis.

Recent developments in business intelligence and artificial intelligence have further improved the way organizations analyze large datasets. AI-assisted tools such as chatbots and AI productivity assistants can help users generate summaries, validate formulas, and provide insights from datasets. These technologies make data analysis faster and more efficient, especially for users who may not have advanced programming or data science skills.

Overall, existing work highlights the importance of structured data management, data cleaning techniques, and the use of analytical tools in improving the accuracy and usefulness of organizational datasets such as employee payroll data.

2.2 Tools and Technologies Review

➤ Microsoft Excel
Microsoft Excel is a spreadsheet software developed by Microsoft and widely used for data management and analysis. It provides features such as formulas, pivot tables, sorting, filtering, and charts that help users analyze datasets efficiently. In this project, Excel is used for data cleaning, payroll calculations, and performing analytical operations on the employee payroll dataset.

➤ Microsoft Word
Microsoft Word is a document processing software used for creating and formatting professional documents. It provides tools for text formatting, inserting tables, adding references, and organizing structured reports. In this project, Microsoft Word is used to prepare the final project report and documentation.

➤ AI Chatbots
AI chatbots are artificial intelligence tools designed to assist users by generating information, answering questions, and supporting analytical tasks. They can help in understanding analytical methods, validating calculations, and generating structured explanations. In this project, AI chatbots are used to support research, explanation of analytical concepts, and documentation assistance.

➤ Microsoft Copilot
Microsoft Copilot is an AI-powered assistant integrated into Microsoft tools such as Excel and Word. It helps automate tasks, suggest formulas, summarize information, and assist with data analysis. In this project, Copilot can be used to support data analysis tasks and improve productivity when working with datasets and reports.

➤ Microsoft Power BI
Power BI is a business intelligence and data visualization tool developed by Microsoft. It enables users to transform raw data into interactive dashboards and visual reports. Power BI helps organizations analyze large datasets and identify meaningful patterns and insights through visual analytics.

2.3 Gaps Identified

1 Limited Analytical Use of Payroll Data

In many organizations, payroll data is primarily maintained for administrative and record-keeping purposes. However, detailed analytical evaluation of payroll datasets is often not performed. As a result, organizations may fail to identify important insights such as salary distribution patterns, departmental payroll costs, and compensation trends.

2 Data Quality and Consistency Issues

Payroll datasets frequently contain issues such as missing values, duplicate records, and inconsistent formatting. Without proper data cleaning and validation processes, these inconsistencies can affect the accuracy of payroll calculations and lead to unreliable analytical results.

3 Manual Processing and Increased Error Risk

Many small and medium-sized organizations still rely on manual spreadsheet operations for payroll analysis. Manual calculations increase the possibility of human errors, especially when handling large datasets, and may reduce overall efficiency in payroll management.

4 Lack of Structured Data Analysis Approaches

There is often a lack of structured analytical approaches that help convert raw payroll data into meaningful insights. Without proper analysis techniques, it becomes difficult to evaluate factors such as average salary trends, department-wise payroll comparisons, and employee compensation variations.

5 Limited Accessibility of Advanced Analytical Tools

Although advanced business intelligence tools are available for data analysis, many users lack the technical expertise required to implement them effectively. Therefore, there is a need for simpler and more accessible methods that utilize widely available tools such as spreadsheets combined with AI-assisted support for data analysis and reporting.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The system design for the payroll data analysis project focuses on transforming raw payroll data into meaningful insights through data cleaning, processing, and visualization. The system is designed to ensure that payroll data can be organized, analyzed, and presented effectively for better understanding and decision-making.

3.1.1 Data Collection

The first step in the system design is collecting the payroll dataset. The dataset contains employee-related information such as employee name, department, gender, salary, bonus, and deductions. This data serves as the primary input for the analysis process.

3.1.2 Data Cleaning and Preparation

Before performing analysis, the dataset is cleaned to remove inconsistencies. This includes handling missing values, correcting formatting issues, and ensuring that numerical data such as salary and bonus fields are properly formatted. Data cleaning improves the accuracy and reliability of the analysis.

3.1.3 Data Processing and Analysis

After cleaning the dataset, analytical techniques are applied to extract meaningful insights. Spreadsheet formulas and pivot tables are used to calculate key metrics such as total payroll cost, average salary, department-wise salary distribution, and employee count per department.

3.1.4 Data Visualization

The analyzed data is then presented using charts and graphs to make the insights easier to understand. Visualizations such as bar charts, pie charts, and column charts are used to represent salary distribution, department-wise payroll costs, and employee statistics.

3.1.5 Dashboard Generation

A dashboard is created to display the summarized insights in a structured format. The dashboard includes key performance indicators such as total employees, total payroll cost, average salary, and highest salary. Interactive charts allow users to quickly understand payroll patterns and trends.

3.1.6 Report Preparation

The final step involves documenting the analysis results and insights. The report presents the methodology, findings, and conclusions derived from the payroll dataset analysis.

3.2 Modules / Components Used

The payroll data analysis system is divided into several modules that work together to process, analyze, and present payroll data effectively. Each module performs a specific function in the overall workflow of the system.

3.2.1 Data Input Module

The data input module is responsible for importing the payroll dataset into the analysis environment. The dataset contains employee-related information such as employee name, department, gender, salary, bonus, and deductions. The data is loaded into the spreadsheet system to begin the analysis process.

3.2.2 Data Cleaning Module

This module focuses on preparing the dataset for analysis by identifying and correcting errors. It includes tasks such as removing duplicate entries, handling missing values, correcting inconsistent formatting, and ensuring that numerical values such as salary and bonus are properly formatted. Proper data cleaning improves the accuracy and reliability of the analysis.

3.2.3 Data Processing Module

The data processing module performs calculations and transformations required for analysis. Spreadsheet formulas and pivot tables are used to compute key metrics such as total payroll cost, average salary, department-wise payroll distribution, and employee count. This module converts raw data into structured information that can be analyzed further.

3.2.4 Data Visualization Module

This module converts processed data into visual representations such as charts and graphs. Visualizations such as bar charts, column charts, and pie charts are used to display payroll insights in an easy-to-understand format. These visuals help users quickly identify patterns and trends within the payroll dataset.

3.2.5 Dashboard Module

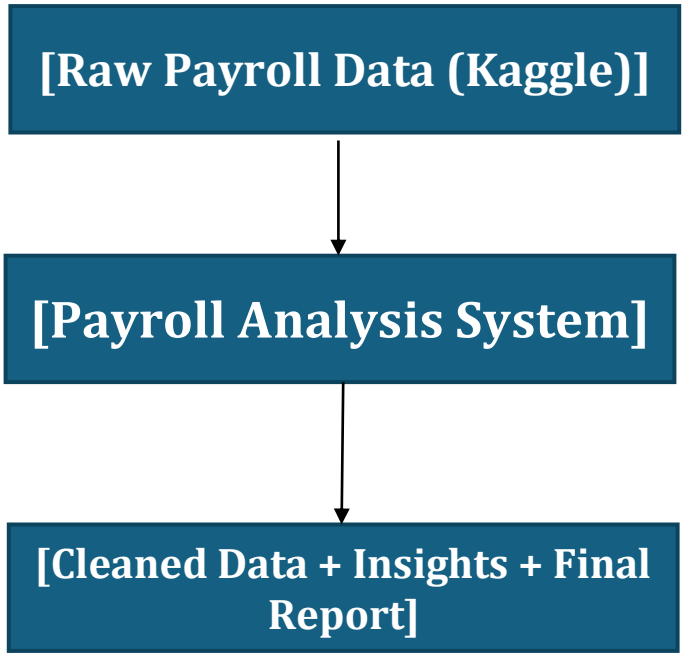
The dashboard module integrates the results of the analysis into a single interactive interface. It displays key performance indicators such as total employees, total payroll cost, average salary, and highest salary along with supporting charts. The dashboard allows users to explore payroll insights efficiently.

3.2.6 Reporting Module

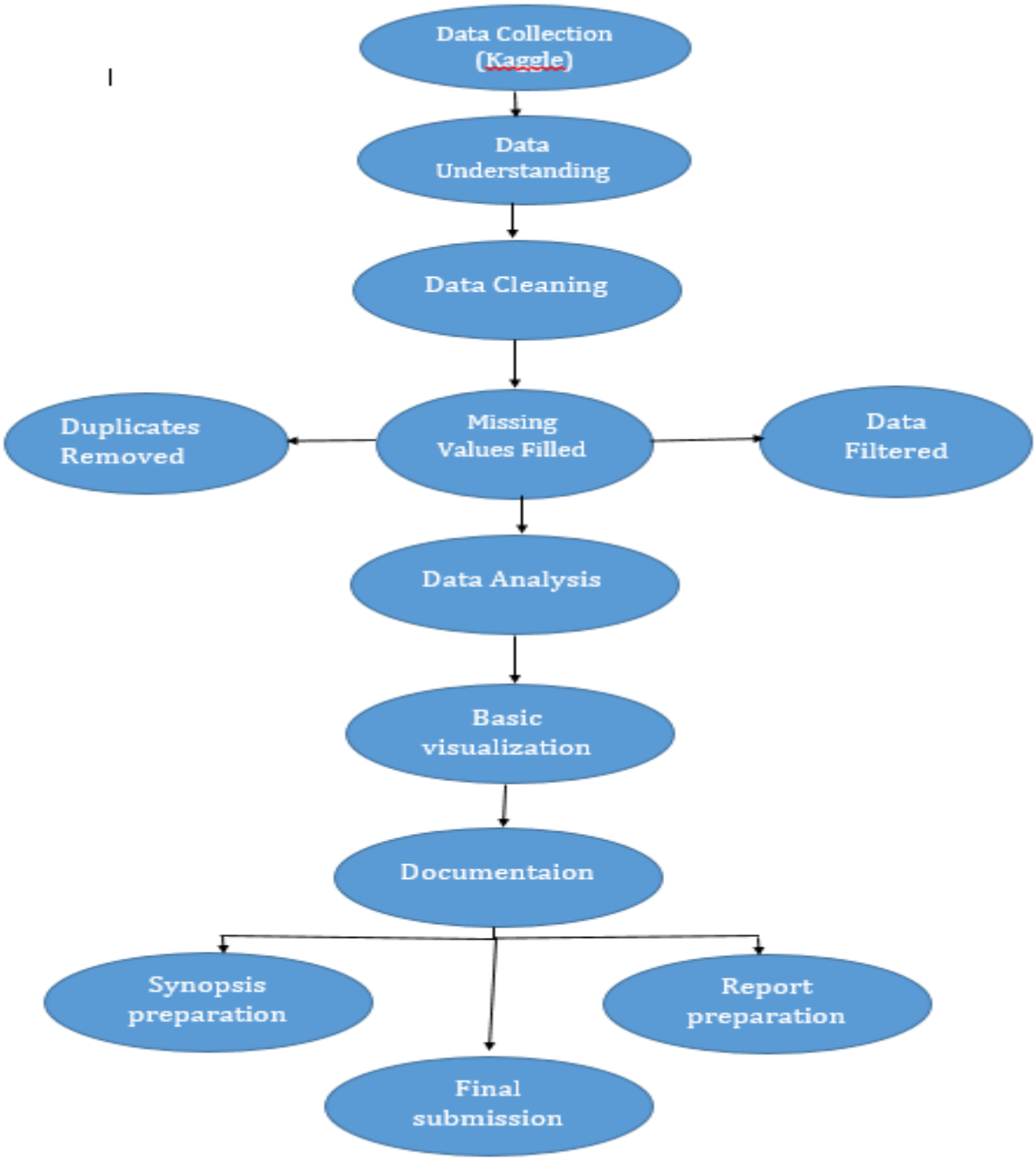
The reporting module documents the results of the analysis in a structured format. It includes explanations of the methodology, visual results, and insights derived from the payroll dataset. The final report provides a comprehensive summary of the payroll data analysis process and findings.

3.3 Data Flow Diagram (DFD)

DFD Level 0 — Context Diagram



DFD Level 1 — Detailed Process Flow



3.4 Advantages

- Reduces manual effort by systematically cleaning payroll data, including removal of duplicates and handling missing values.
- Improves accuracy of salary calculations, which helps in minimizing payroll errors and building employee trust.
- Provides clear and structured insights such as total payroll cost and department-wise distribution, making it easy for management to understand.
- Supports better decision-making by transforming raw payroll data into meaningful and actionable information.
- Uses AI tools like ChatGPT or Copilot to assist in validation, error detection, and generating summaries, which reduces human error and speeds up the process.
- Scalable approach, as the same process can be applied to larger datasets or other business domains like sales or finance.
- Cost-effective, as it uses commonly available tools like Excel without requiring advanced or expensive software.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

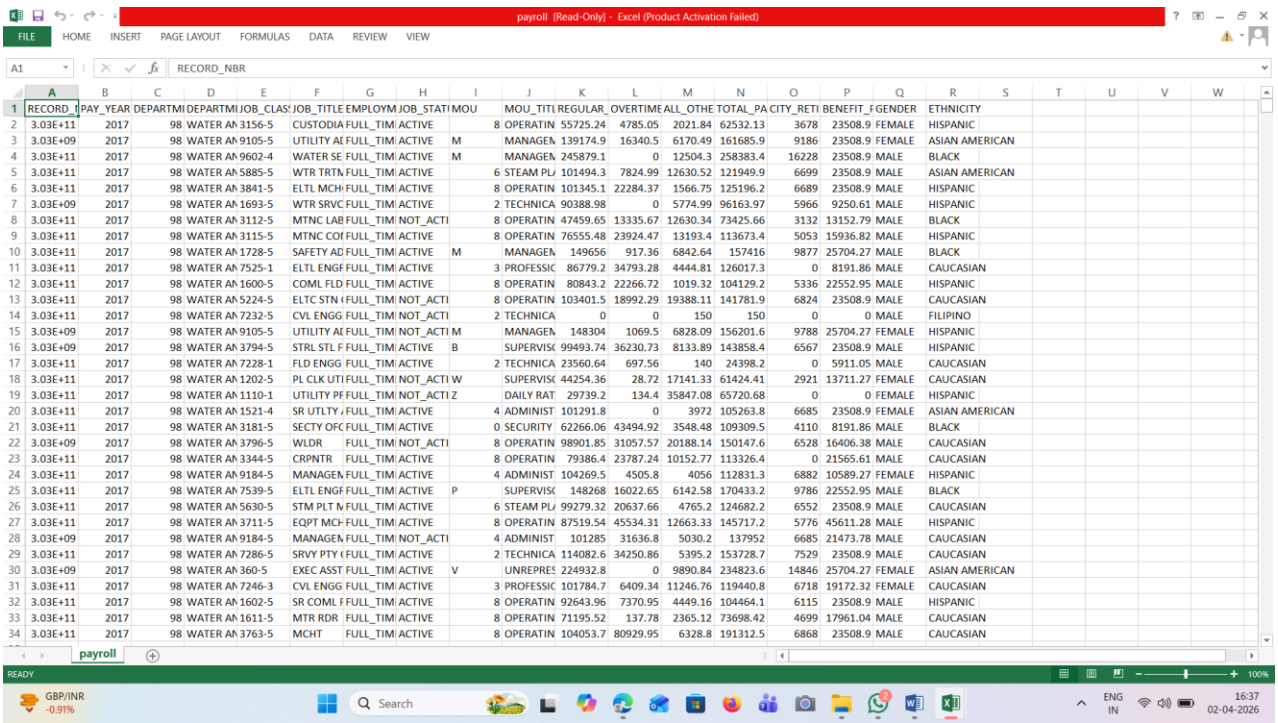
CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

- Dataset source: https://www.kaggle.com/datasets/dsfelix/employees-payroll-in-los-angeles
- Original dataset size: Number of rows : 12329229 ; Number of columns : 685462
- Issues found: Duplicates, null values, wrong data types, formatting inconsistencies
- Cleaning steps applied: Removed duplicates, dealed with missing values, corrected the format, added necessary row/column.
- Cleaned dataset size: number of row : 1595347 ; number of column : 65536



RECORD_	PAY_YEAR	DEPARTMI	DEPARTMI	JOB_CLAS	JOB_TITLE	EMPLOYM	JOB_STATI	MOU	MOU_TITI	REGULAR_	OVERTIME	ALL_OTHE	TOTAL_PA	CITY_RETI	BENEFIT_F	GENDER	ETHNICITY
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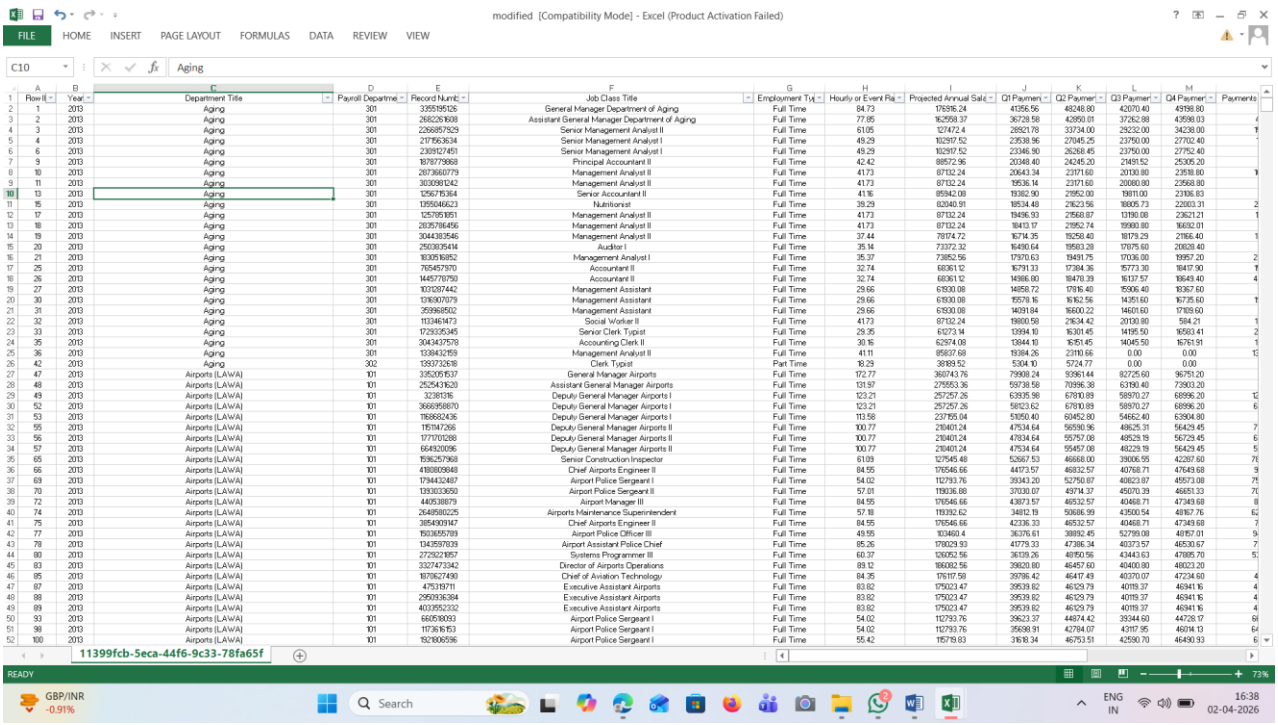
Uncleaned Dataset



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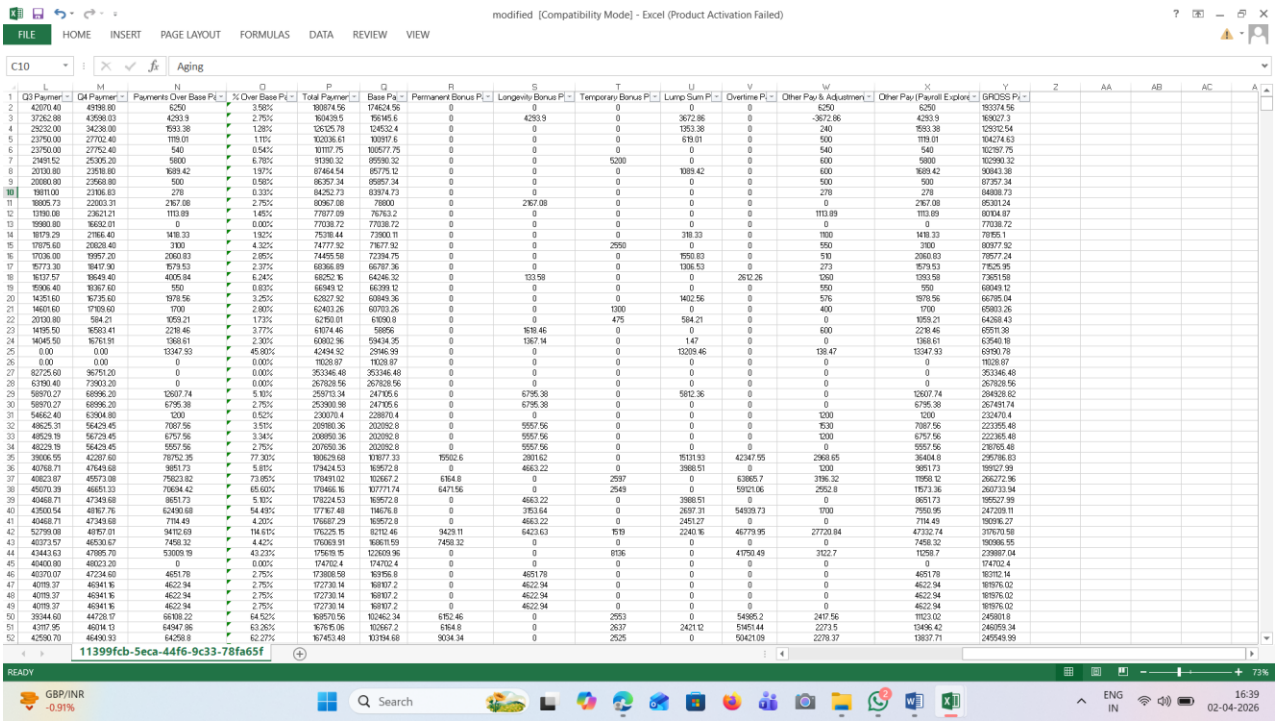
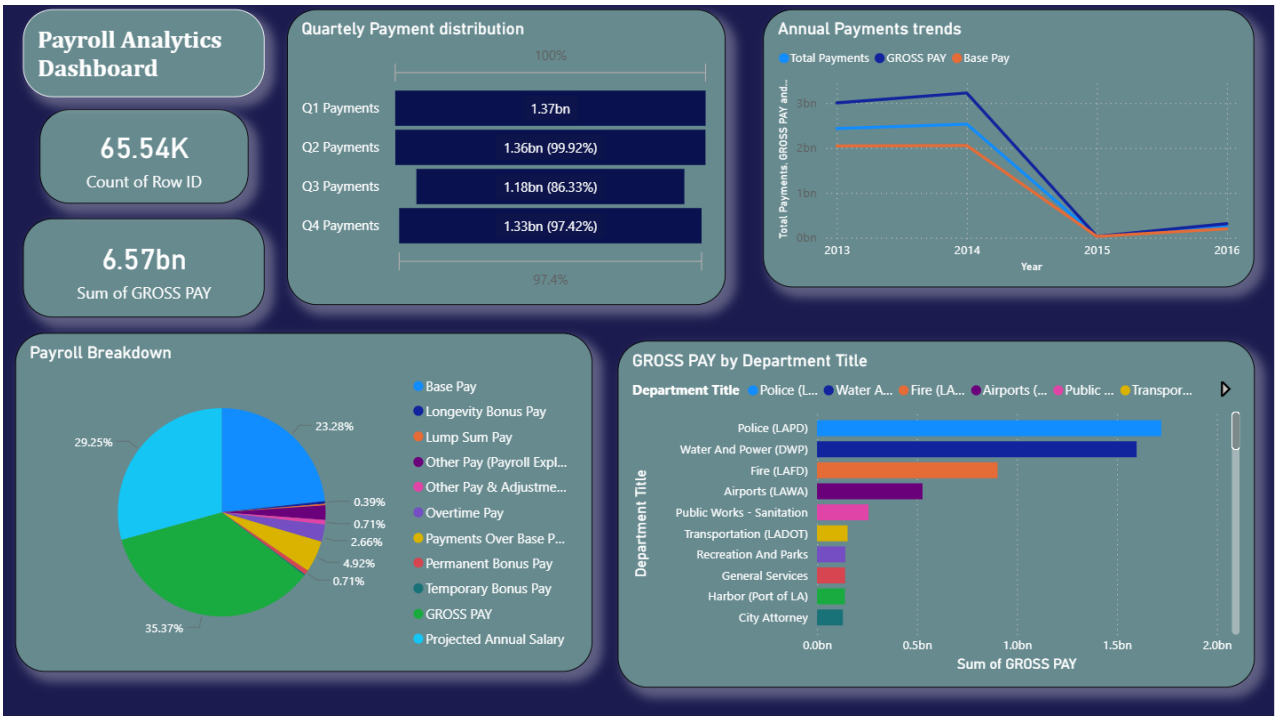
2	2013	Aging	301	2662051608	Assistant General Manager Department of Aging	Full Time	77.85	162558.37	36728.58	42850.01	37362.88	43918.03	4
3	2013	Aging	301	2266857529	Senior Management Analyst II	Full Time	61.05	127472.4	28921.78	33734.00	29232.00	34238.00	1
4	2013	Aging	301	2117563634	Senior Management Analyst I	Full Time	48.29	102917.52	23538.96	27045.25	23750.00	27702.40	1
5	2013	Aging	301	2339127451	Senior Management Analyst I	Full Time	48.29	102917.52	23346.90	26268.45	23750.00	27752.40	
6	2013	Aging	301	1878773868	Principal Accountant II	Full Time	42.42	88572.96	20348.40	24245.20	21491.52	25305.20	
7	2013	Aging	301	2873660779	Management Analyst II	Full Time	41.73	87132.24	20643.34	23171.60	20130.80	23518.80	1
8	2013	Aging	301	3030991242	Management Analyst II	Full Time	41.73	87132.24	19536.14	23171.60	20880.80	23568.80	
9	2013	Aging	301	1256715364	Senior Accountant II	Full Time	41.16	85942.08	19382.90	21952.00	19811.00	23106.83	
10	2013	Aging	301	1295046623	Nutritionist	Full Time	35.29	63040.91	18534.48	21623.56	18805.73	22003.31	2
11	2013	Aging	301	1257651851	Management Analyst II	Full Time	41.73	87132.24	19496.93	21568.87	13180.08	23621.21	1
12	2013	Aging	301	2835796456	Management Analyst II	Full Time	41.73	87132.24	18413.17	21952.74	19980.80	16692.01	
13	2013	Aging	301	3044383546	Management Analyst II	Full Time	37.44	78174.72	16714.35	19258.40	18179.29	21166.40	1
14	2013	Aging	301	2503035414	Auditor I	Full Time	35.14	73372.32	16490.64	19503.28	17975.60	20828.40	
15	2013	Aging	301	1830516852	Management Analyst I	Full Time	35.37	73852.56	17970.63	19491.75	17036.00	19957.20	2
16	2013	Aging	301	765457970	Accountant II	Full Time	32.74	68361.12	16791.33	17384.36	15773.30	18417.90	1
17	2013	Aging	301	1445776750	Accountant II	Full Time	32.74	68361.12	14986.80	18478.39	16137.57	18649.40	4
18	2013	Aging	301	1031297442	Management Assistant	Full Time	29.66	61930.08	14858.72	17916.40	15906.40	18367.60	
19	2013	Aging	301	1316907079	Management Assistant	Full Time	29.66	61930.08	15578.16	16162.56	14351.60	16735.60	1
20	2013	Aging	301	359968502	Management Assistant	Full Time	29.66	61930.08	14091.84	16600.22	14601.60	17109.60	
21	2013	Aging	301	1113461473	Social Worker II	Full Time	41.73	87132.24	19800.58	21634.42	20130.80	584.21	1
22	2013	Aging	301	1723935345	Senior Clerk Typist	Full Time	29.35	61273.14	13994.10	16301.45	14195.50	16593.41	2
23	2013	Aging	301	3043437578	Accounting Clerk II	Full Time	30.16	62974.08	13844.10	16151.45	14045.50	16761.91	1
24	2013	Aging	301	1339432159	Management Analyst II	Full Time	41.11	85837.68	19384.26	23110.66	0.00	0.00	1
25	2013	Aging	302	1393732618	Clerk Typist	Part Time	18.29	38188.52	5304.10	5724.77	0.00	0.00	
26	2013	Airports (LAWA)	101	3352051537	General Manager Airports	Full Time	172.77	360743.76	79908.24	93961.44	82725.60	96751.20	
27	2013	Airports (LAWA)	101	2525431620	Assistant General Manager Airports	Full Time	131.97	275553.36	59738.58	70996.38	63190.40	73903.20	
28	2013	Airports (LAWA)	101	32381316	Deputy General Manager Airports I	Full Time	123.21	257257.26	63935.98	67810.89	58970.27	68996.20	1
29	2013	Airports (LAWA)	101	3669358670	Deputy General Manager Airports I	Full Time	123.21	257257.26	58123.62	67810.89	58970.27	68996.20	6
30	2013	Airports (LAWA)	101	1868822436	Deputy General Manager Airports I	Full Time	113.58	237155.04	51850.40	60452.80	54662.40	63904.80	
31	2013	Airports (LAWA)	101	1757147266	Deputy General Manager Airports II	Full Time	100.77	210401.24	47534.64	56590.96	48625.31	56429.45	7
32	2013	Airports (LAWA)	101	1771701288	Deputy General Manager Airports II	Full Time	100.77	210401.24	47834.64	55757.08	48529.19	56729.45	6
33	2013	Airports (LAWA)	101	1664320096	Deputy General Manager Airports II	Full Time	100.77	210401.24	47534.64	55457.08	48229.19	56429.45	5
34	2013	Airports (LAWA)	101	1596257968	Senior Construction Inspector	Full Time	61.09	127545.48	52667.53	46668.00	39006.55	42267.60	7
35	2013	Airports (LAWA)	101	4188809848	Chief Airports Engineer II	Full Time	84.55	176546.66	44173.57	46832.57	40768.71	47349.68	9
36	2013	Airports (LAWA)	101	1794432487	Airport Police Sergeant I	Full Time	54.02	112793.76	39343.20	52750.87	40823.87	45573.08	7
37	2013	Airports (LAWA)	101	1393033650	Airport Police Sergeant II	Full Time	57.01	119036.88	37030.07	49714.37	45070.39	46651.33	7
38	2013	Airports (LAWA)	101	440538679	Airport Manager III	Full Time	84.55	176546.66	43873.57	46532.57	40468.71	47349.68	8
39	2013	Airports (LAWA)	101	2648980025	Airports Maintenance Superintendent	Full Time	57.18	119392.62	34812.19	50686.99	43500.54	48167.76	6
40	2013	Airports (LAWA)	101	3854930147	Chief Airports Engineer II	Full Time	84.55	176546.66	42336.33	46532.57	40468.71	47349.68	7
41	2013	Airports (LAWA)	101	1503655789	Airport Police Officer III	Full Time	49.55	103460.4	36376.61	38892.45	52799.08	49157.01	9
42	2013	Airports (LAWA)	101	1343597839	Airport Assistant Police Chief	Full Time	85.26	178029.93	41779.33	47386.34	40373.57	46530.67	7
43	2013	Airports (LAWA)	101	2723021857	Systems Programmer III	Full Time	60.37	126052.56	36139.26	49150.56	43443.63	47885.70	5
44	2013	Airports (LAWA)	101	3327473342	Director of Airports Operations	Full Time	89.12	186082.56	39820.80	46457.60	40400.80	48023.20	
45	2013	Airports (LAWA)	101	1870627490	Chief of Aviation Technology	Full Time	84.35	176117.59	39786.42	46417.49	40370.07	47234.60	4
46	2013	Airports (LAWA)	101	475319711	Executive Assistant Airports	Full Time	83.82	175023.47	39539.82	46129.79	40119.37	46941.16	4
47	2013	Airports (LAWA)	101	2959036384	Executive Assistant Airports	Full Time	83.82	175023.47	39539.82	46129.79	40119.37	46941.16	4
48	2013	Airports (LAWA)	101	4033552332	Executive Assistant Airports	Full Time	83.82	175023.47	39539.82	46129.79	40119.37	46941.16	4
49	2013	Airports (LAWA)	101	660518093	Airport Police Sergeant I	Full Time	54.02	112793.76	39623.37	44874.42	39344.60	44728.17	6
50	2013	Airports (LAWA)	101	1173616153	Airport Police Sergeant I	Full Time	54.02	112793.76	35938.91	42784.07	43117.95	46014.13	6
51	2013	Airports (LAWA)	101	1921806596	Airport Police Sergeant I	Full Time	55.42	115718.83	31618.34	46753.51	42590.70	46490.93	6

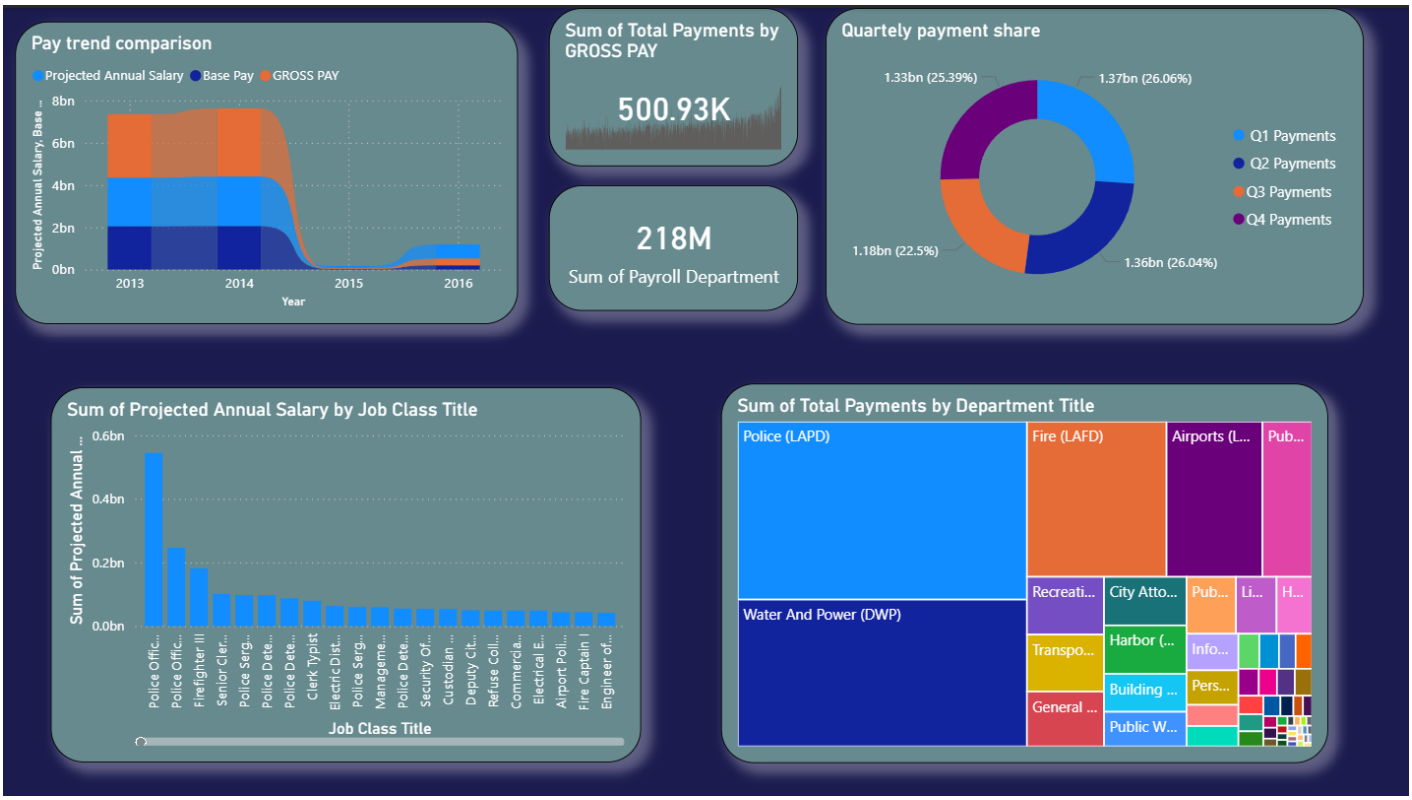
Cleaned Dataset



Q3 Paymen	Q4 Paymen	Payments Over Base Pi	% Over Base Pi	Total Paymen	Base Pa	Permanent Bonus Pi	Longevity Bonus Pi	Temporary Bonus Pi	Lump Sum Pi	Overtime Pi	Other Pay & Adjustmen	Other Pay (Payroll Explore	GROSS Pi
42070.40	49918.80	6250	3.56%	180874.56	174624.56	0	0	0	0	0	6250	6250	193374.56
37262.88	43598.03	4293.9	2.75%	160439.5	156145.6	0	4293.9	0	3672.86	0	-3672.86	4293.9	168027.3
29232.00	34238.00	1593.38	1.28%	126125.78	124532.4	0	0	0	1353.38	0	240	1593.38	129312.54
23750.00	27702.40	1119.01	1.11%	102036.61	100917.6	0	0	0	619.01	0	500	1119.01	104274.63
23750.00	27752.40	540	0.54%	101197.75	100577.75	0	0	0	0	0	540	540	102297.75
21491.52	25305.20	5800	6.78%	91390.32	85590.32	0	0	5200	0	0	600	5800	102990.32
20130.80	23518.80	1689.42	1.97%	87464.54	85775.12	0	0	0	1089.42	0	600	1689.42	90843.38
20880.80	23568.80	500	0.58%	86357.34	85857.34	0	0	0	0	0	500	500	87357.34
19811.00	23106.83	278	0.33%	84252.73	83974.73	0	0	0	0	0	278	278	84808.73
18805.73	22003.31	2167.08	2.75%	80967.08	78800	0	2167.08	0	0	0	0	2167.08	85301.24
13180.08	23621.21	1113.89	1.45%	77877.69	76763.2	0	0	0	0	0	1113.89	1113.89	80104.87
19980.80	16692.01	0	0.00%	77038.72	77038.72	0	0	0	0	0	0	0	77038.72
18179.29	21166.40	1418.33	1.92%	75318.44	73900.11	0	0	0	318.33	0	1100	1418.33	78955.1
17875.60	20828.40	3100	4.32%	74777.92	71677.92	0	0	2550	0	0	550	3100	80977.92
17036.00	19957.20	2060.83	2.95%	74455.58	72394.75	0	0	0	1550.83	0	510	2060.83	76577.24
15773.30	18417.90	1579.53	2.37%	68366.89	66787.36	0	0	0	1306.53	0	273	1579.53	71525.95
16137.57	18649.40	4005.94	6.24%	68252.16	64246.32	0	123.58	0	0	2612.26	1293	4005.94	73651.58
15906.40	18367.60	550	0.83%	66949.12	66399.12	0	0	0	0	0	550	550	68049.12
14351.60	16735.60	1978.56	3.25%	62827.92	60849.36	0	0	0	1402.56	0	576	1978.56	66785.04
14601.60	17109.60	1700	2.80%	62403.26	60703.26	0	1300	0	0	0	400	1700	65803.26
20130.80	584.21	1059.21	1.73%	62150.01	61090.8	0	475	584.21	0	0	0	1059.21	64268.43
14195.50	16593.41	2218.46	3.77%	61074.46	58856	0	1618.46	0	0	0	600	2218.46	65511.38
14045.50	16761.91	1368.61	2.30%	60802.96	59434.35	0	1367.14	0	1.47	0	0	1368.61	63540.18
0.00	0.00	13347.93	45.80%	42494.92	29146.99	0	0	0	13209.46	0	138.47	13347.93	63180.78
0.00	0.00	0	0.00%	11028.87	11028.87	0	0	0	0	0	0	0	11028.87
82725.60	96751.20	0	0.00%	353346.48	353346.48	0	0	0	0	0	0	0	353346.48
63190.40	73903.20	0	0.00%	267828.56	267828.56	0	0	0	0	0	0	0	267828.56
58970.27	68996.20	12607.74	5.10%	259713.34	247105.6	0	6795.38	0	5812.36	0	0	12607.74	264328.62
58970.27	68996.20	6795.38	2.75%	253900.98	247105.6	0	6795.38	0	0	0	0	6795.38	267481.74
54662.40	63904.80	1200	0.52%	230070.4	228870.4	0	0	0	0	0	1200	1200	232470.4
48625.31	56429.45	7087.56	3.51%	209180.36	202092.8	0	5557.56	0	0	0	1530	7087.56	223355.48
48529.19	56729.45	6757.56	3.34%	208950.36	202092.8	0	5557.56	0	0	0	1200	6757.56	222365.48
48229.19	56429.45	5557.56	2.75%	207650.36	202092.8	0	5557.56	0	0	0	0	5557.56	218765.48
39006.55	42267.60	78752.35	77.30%	180629.68	101877.33	15592.6	2801.62	0	15131.93	42347.55	2968.65	36404.8	295786.63
40768.71	47649.68	9851.73	5.81%	179424.53	169572.8	0	4663.22	0	3988.51	0	1200	9851.73	199127.99
40823.87	45573.08	75823.82	73.85%	178491.02	102667.2	6164.8	0	2597	0	63865.7	3196.32	11958.12	266272.96
45070.39	46651.33	70694.42	65.60%	178466.16	107771.74	6471.56	0	2549	0	59121.06	2552.8	11573.36	260733.94
40468.71	47349.68	8851.73	5.10%	178224.53	169572.8	0	4663.22	0	3988.51	0	0	8851.73	195527.99
43500.54	48167.76	62490.68	54.49%	177167.48	114676.8	0	3153.64	0	2697.31	54939.73	1700	7550.95	247209.11
40468.71	47349.68	7114.49	4.20%	176687.29	169572.8	0	4663.22	0	2451.27	0	0	7114.49	190916.27
52799.08	49157.01	94112.69	114.61%	176225.15	82112.46	9429.11	6423.63	1519	2240.16	46779.95	27720.84	47332.74	317670.58
40373.57	46530.67	7458.32	4.42%	176063.91	168611.59	7458.32	0	0	0	0	7458.32	7458.32	190986.55
43443.63	47885.70	53009.19	43.23%	175619.15	122609.96	0	0	8136	0	41750.49	3122.7	11258.7	239897.04
40400.80	48023.20	0	0.00%	174702.4	174702.4	0	0	0	0	0	0	0	174702.4
40370.07	47234.60	4851.78	2.75%	173808.58	168956.8	0	4851.78	0	0	0	0	4851.78	163112.14
40119.37	46941.16	4622.94	2.75%	172730.14	168107.2	0	4622.94	0	0	0	0	4622.94	161976.02
40119.37	46941.16	4622.94	2.75%	172730.14	168107.2	0	4622.94	0	0	0	0	4622.94	161976.02
40119.37	46941.16	4622.94	2.75%	172730.14	168107.2	0	4622.94	0	0	0	0	4622.94	161976.02
39344.60	44728.17	66108.22	64.52%	168570.56	102462.34	6152.46	0	2593	0	54985.2	2417.56	11123.02	245801.8
43117.95	46014.13	64947.86	63.26%	167675.06	102667.2	6164.8	0	2637	2421.12	51451.44	2273.5	13496.42	246893.34
42590.70	46490.93	64258.8	62.27%	167453.48	103194.68	9034.34	0	2525	0	50421.09	2278.37	13837.71	245549.99

4.2 Visualization / Output Results





4.3 AI Tool Integration Results

Describe how the AI tool (Copilot or Chatbot) was used in your project and what results it produced.

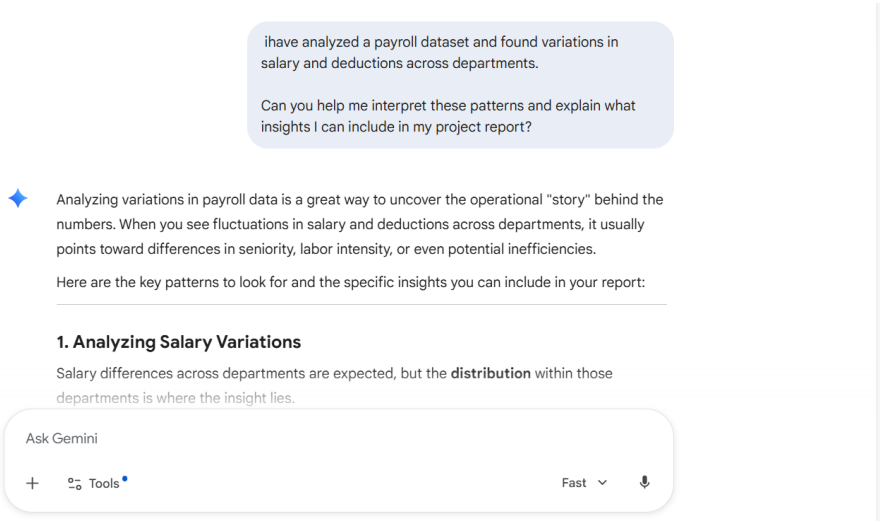
- Tool used: Microsoft Copilot , ChatGPT , Google Gemini
- Prompts used: Assist in optimizing Excel formulas for large datasets and improving efficiency in calculations.Help generate structured summaries for department-wise payroll analysis.Provide insights based on trends observed in the data.

- Output quality: The AI-generated outputs were clear, relevant, and context-specific, especially when dealing with complex aspects of payroll analysis. The suggestions provided for validating salary calculations, identifying inconsistencies, and interpreting patterns helped improve the accuracy of the analysis.

  AI tools were particularly useful in refining insights, structuring explanations, and enhancing the overall presentation of the report. However, all outputs were carefully reviewed and validated before implementation to ensure correctness and alignment with the dataset.

  Overall, the use of AI tools improved efficiency, reduced manual effort in complex tasks, and supported better understanding of analytical results.

- How it enhanced the project: The use of AI tools enhanced the project by improving both the speed and quality of complex analysis tasks, such as validating payroll calculations and identifying inconsistencies in large datasets. It helped in generating more accurate insights and provided better ways to interpret patterns like salary variations and deduction trends. Additionally, AI supported in refining explanations and structuring the report more effectively, making the overall analysis clearer, more efficient, and professionally presented.



I have a payroll dataset in Excel with columns like Gross Salary, Deductions, and Net Salary.
Can you help me check if Net Salary is correctly calculated and suggest a formula to identify errors in large datasets?

Yes, you can validate this easily in Excel.

First, the expected relationship is:
Net Salary = Gross Salary – Deductions

To check this, create a new column and use:

```
=IF(C2-B2=D2,"Correct","Error")
```

(Assuming Gross Salary = C2, Deductions = B2, Net Salary = D2 — adjust as per your sheet)

For better accuracy (handling rounding differences), use:

```
=IF(ABS((C2-B2)-D2)<1,"Correct","Error")
```

Then apply the formula to all rows.
It will flag records as **"Correct"** or **"Error"**, helping you quickly identify issues in large datasets.

Ask anything

Optimizing Excel for Payroll Analysis

Help me optimize Excel formulas for analyzing a large payroll dataset and suggest efficient ways to calculate total payroll cost and department-wise salary distribution.

To optimize payroll analysis in Excel, use structured formulas like `SUMIFS` for department-wise totals and `SUMPRODUCT` for overall payroll cost. Combine these with PivotTables for scalability, ensuring fast calculations even with large datasets.

🔑 Key Formulas for Payroll Analysis

1. Total Payroll Cost

- **Formula:**

Excel Copy

```
=SUMPRODUCT(SalaryRange, HoursWorkedRange)
```

- Efficient for large datasets since it multiplies salary by hours worked across rows without needing helper columns.

Ask anything

Smart

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- Key Finding 1: After cleaning, 1,00,000 + of the records were found to be duplicates — removing them significantly improved dashboard accuracy.]
- Key Finding 2: The AI Copilot reduced the time needed to generate a summary report from 2 hours to under 10 minutes.

5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project: [Paste your GitHub repository link here]
- Video Recording of Project Demonstration: [Paste your YouTube / Google Drive / OneDrive video link here]
- Power BI / Dashboard Link (if applicable): [Paste published dashboard link here]

5.3 Limitations

- Limitation 1: The project was tested on the city dataset and not company dataset.
- Limitation 2: The analysis was performed using Excel, which has limitations in handling extremely large datasets efficiently compared to advanced tools like databases or Python.
- Limitation 3: Due to extreme large data set understanding dashboard becomes hactic.

5.4 Future Work

- The project can be extended by using a real-time payroll dataset from an organization so that the analysis reflects current salary structures and business conditions.
- More advanced tools such as Power BI, SQL, or Python can be integrated to handle larger datasets and generate deeper insights more efficiently.
- Machine learning techniques can be applied in the future to detect unusual salary patterns, predict payroll expenses, or identify possible errors automatically.
- An interactive dashboard or web-based interface can be developed so that HR and finance teams can easily view payroll insights without manually using Excel.
- Additional payroll-related factors such as employee attendance, overtime hours, leave records, and performance-based incentives can be included for more detailed analysis.

5.5 Conclusion

This project addressed the challenge of managing and analyzing a large and complex payroll dataset. Through systematic data cleaning and preprocessing, the dataset was transformed into a structured and reliable format suitable for analysis. Using Excel-based techniques, key insights such as total payroll cost, salary distribution, and validation of net salary calculations were successfully derived.

All the defined objectives were achieved, including data cleaning, KPI analysis, and integration of AI tools for validation and insight generation. The use of AI enhanced efficiency and supported better interpretation of results while ensuring that core analytical tasks were performed independently.

Overall, the project demonstrates how structured data analysis combined with selective use of AI tools can improve accuracy, efficiency, and decision-making in payroll management. This fellowship experience helped in developing practical skills in data analysis, problem-solving, and professional reporting.

REFERENCES

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com

[4] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

https://www.kaggle.com/datasets/dsfelix/employees-payroll-in-los-angeles

Appendix B — Key Excel Formulas / DAX Measures Used

SUM – Used to calculate total payroll cost

AVERAGE – Used to find average salary

MAX – Used to identify highest salary

MIN – Used to identify lowest salary

COUNTIF – Used to count duplicate or specific records

SUMIF – Used for department-wise salary calculation

IF – Used for validation (Correct/Error check)

IFERROR – Used to handle errors in formulas

Appendix C — AI Prompts Used

Help validate payroll calculations in my dataset, especially checking whether Net Salary is correctly calculated from Gross Salary and deductions.

Suggest efficient Excel formulas to identify errors and inconsistencies in a large payroll dataset.

Help interpret payroll analysis results such as salary distribution and deduction patterns for reporting.

Assist in structuring a clear and professional summary of payroll insights for my project report.

Appendix D — Screenshot Gallery

All inluded